



Macronutrients

LESSON 4

What are Macronutrients?

Nutrients we need in large amounts:

- Carbohydrate
- Fat
- Protein

We use them for fuel, building muscle and to provide other nutrients like fibre, vitamins & minerals



Carbohydrates

Carbohydrate foods:

- Bread
- Pasta
- Rice
- Potatoes
- Cereals
- Grains
- Sugars like table sugar and honey are also carbohydrates.
- Carbohydrates are the main fuel for our bodies.
- All carbs are broken down to glucose during digestion.
- Muscles, kidneys, brain, heart and nervous system all need glucose to function.



Did you know your brain uses 120g of glucose everyday? That's almost $\frac{1}{4}$ of your daily calories.

Carbohydrate & Sports

- Used by the body for fuel.
- Muscles store carbs as glycogen.
- Glycogen is used for fuel during exercise.
- Elite athletes often need more carbs than everyone else.



How much fuel do I need?

Gender	Calories Inactive	Calories Active
Girls	1800 kcal	2000 kcal
Boys	2200 kcal	2400 -2800 kcal

- It is best to get most of your fuel from wholegrain and high fibre carbohydrate foods.
- These foods bring with them other nutrients like fibre, iron and B vitamins.



Carbohydrate & Fibre

- Carb foods are a good source of fibre – as long as you choose wholegrain not refined options.
- Wholegrain bread
- Brown Rice & pasta
- Jacket Potatoes
- High fibre cereals

Benefits of fibre:

- Reduces heart disease, diabetes & cancer.
- Helps to maintain a healthy weight.
- We need 25g of fibre per day.
- Check the label: 6g of fibre or more per 100g is high fibre.



Where else can I get fibre?

Food	g of Fibre
Slice of wholegrain bread	2 - 3g
Jacket potato with skin	3g
Bowl of high fibre cereal	3 - 4g
1 dessertspoon seeds	3g
½ tin baked beans	7g
Small tin chickpeas	10g
Handful of nuts	3 - 4g
Fruit or 1 serving of vegetables	2 - 3g

Carbohydrate & Glycaemic Index

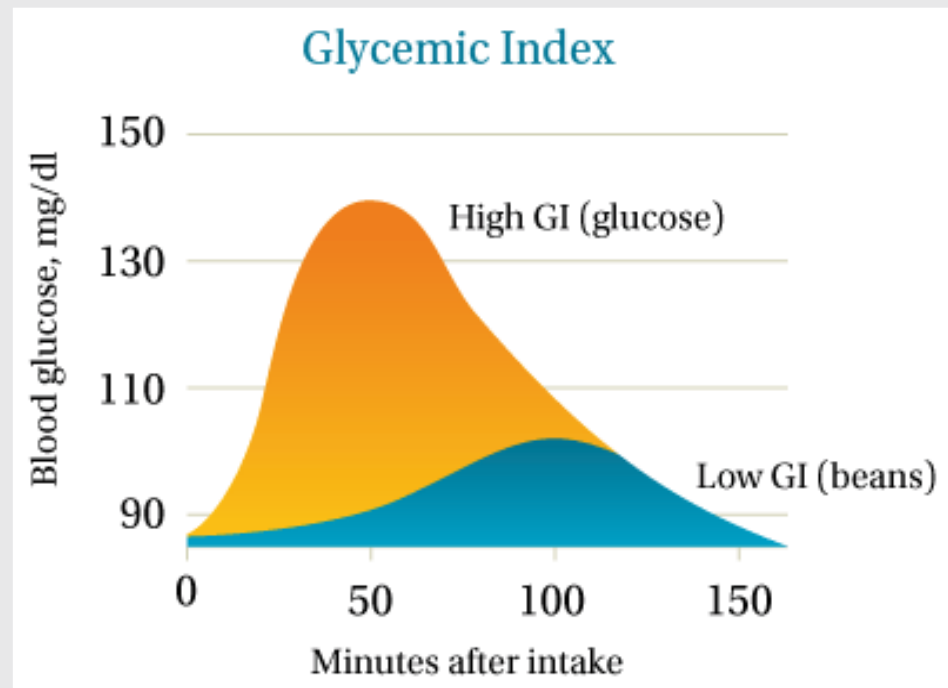
- **Glycaemic Index** measures how high your blood sugar goes after eating a food.
- A lower, steady blood sugar helps maintain energy levels and concentration.

Low GI foods:

- Brown rice, Pasta,
- Porridge, Fruit & vegetables, Beans & lentils

High GI foods:

- Sugars, white rice, potatoes eaten without skins



Protein

Protein is needed for:

- Muscles
- Bones
- Skin & Hair
- Hormones
- Enzymes
- Immune system
- Can be used as fuel.
- Did you know your body does not store protein?
- Your body takes the protein it needs from your food.
- Any protein left over is used for fuel or may be stored as fat.



Protein & Health



Protein Foods are also good sources of other nutrients including:

- Iron
- Zinc
- Selenium
- B Vitamins

Protein food include:

- Meat, Chicken, Fish
- Eggs, Beans, Nuts
- We need protein foods at two meals per day

Fat

Fat is needed for:

- Healthy growth & development
- Hormones
- Skin & hair
- Healthy Brain
- Fuel
- All cell membranes need fat

Fat is very high in calories:

- 1g of carbs had 4kcal
- 1g of fat has 9kcal
- Do use some fat but keep it to small amounts.



Types of Fat

Saturated fats

- Less healthy type of fat.
- Linked with heart disease, diabetes and some cancers.
- Foods high in saturated fat are often high in salt and sugar as well.
- Processed meats (sausages, pudding etc.); cakes, biscuits, pastries, cream, butter.



Unsaturated fats

- These are healthy fats.
- Linked with a reduced risk of heart disease.
- Good source of vitamin E.
- Can be good sources of omega-3 fats.
- Olive oil, rapeseed oil, sunflower oil, nuts, seeds.



Omega-3 Fats

- The human brain is almost 60% of fat.
- Most of this fat is DHA, a type of omega-3 fat.
- DHA is found in oil-rich fish.
- ~~Salmon~~ Salmon, trout, mackerel, herring, sardines, fresh
- Eat oil rich fish at least once a week.



Balance Your Plate

Carbs

- Pasta, Potatoes, Wholegrain, Bread

Vegetables

- Vegetables & Salad

Protein

- Meat, Fish, Chicken, Beans

Healthy fats can be included in cooking & on salads.

